CASE STUDY — Titan Manufacturing Corporation – The Agentic AI Challenge

Background

Titan Manufacturing Corporation (TMC) is a global leader in industrial machinery, robotics components, and precision equipment, with 56,000 employees and operations across 14 countries. Titan runs 28 manufacturing plants, each mixing advanced robotics with aging operational technology (OT).

Despite heavy investments in automation and IoT sensors, Titan struggles with **unplanned downtime**, supply chain volatility, and plant-floor decision bottlenecks. Leaders believe Agentic AI can unify their OT and IT environments, enabling predictive operations and autonomous coordination.

The Scenario: A Constellation of Problems Under One Root Cause

Titan's production throughput has declined by 9% year-over-year. Customer delivery SLAs are frequently breached due to delayed materials, unexpected machine failures, and operator-robot coordination issues.

While each team blames different factors—maintenance blames procurement, procurement blames logistics, engineering blames outdated PLCs—executives see a common pattern:

Titan suffers from fragmented operational intelligence, isolated systems, and entirely manual decision loops.

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1. Asset Reliability & Predictive Maintenance Challenges

Symptoms

- Plants record **22,000+ sensor alerts daily**, with no prioritization.
- Mean time between failures (MTBF) dropped **11%** last year.
- Critical CNC machines cause **$180,000/day** losses when down.
- 38% of downtime tickets indicate issues that "should have been predicted."

Underlying Issues

- Sensor data lives across SCADA, historians, PLC logs, and vendor portals.
- No Remaining Useful Life (RUL) predictions.
- Maintenance scheduling remains rule-based instead of data-driven.
- Human technicians cannot review raw telemetry at scale.

Agentic AI Opportunities to be identified

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2. Supply Chain Volatility & Risk Challenges

Symptoms

- Supplier delivery delays increased by **28%**.
- Line stoppages due to missing parts cost **$14M** last quarter.
- Tier-2 supplier issues uncovered only *after* production stalls.
- Expedited logistics expenses up **52%**.

Underlying Issues

- No visibility beyond Tier-1 suppliers.
- Logistics data spread across email, EDI feeds, and spreadsheets.
- No real-time ETA predictions.

Agentic AI Opportunities to be identified

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3. Human–Robot Coordination Challenges

Symptoms

- 17% of production shifts encounter robot-human scheduling conflicts.
- Operators frequently override robot sequences, causing quality inconsistencies.
- Safety shutdowns increased **15%** last year.
- Robot cells cannot dynamically adjust to staffing changes.

Underlying Issues

- Schedulers operate offline and manually.
- No real-time safety context for robots.
- Human intention is not captured for dynamic adjustment.

- No unified coordination engine.

Agentic AI Opportunities to be identified

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4. Quality & Traceability Challenges

Symptoms

- Quality escapes increased **22%** YTD.
- Defects traced to upstream steps take **weeks** to identify.
- Plant quality processes vary broadly, creating audit inconsistencies.

Underlying Issues

- Quality data in MES/QMS not integrated with robotics telemetry.
- Root-cause analysis depends on tribal knowledge.

Agentic AI Opportunities to be identified

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5. Compliance, Safety & Audit Challenges

Symptoms

- OSHA compliance gaps due to incomplete digital logs.
- Audit cycles require manual cross-referencing of 20+ systems.
- Safety event investigations take too long to resolve.

Underlying Issues

- OT and safety systems are entirely siloed.
- No automated incident reconstruction.

Agentic AI Opportunities to be identified